

Task 7

Input:

```
import sqlite3

import pandas as pd

import matplotlib.pyplot as plt


# Connect to SQLite (in-memory for example)
conn = sqlite3.connect(":memory:")
cursor = conn.cursor()


# Create the sales table with 10 columns
cursor.execute("""
CREATE TABLE sales (
    sale_id INTEGER PRIMARY KEY,
    product_id INTEGER,
    product_name TEXT,
    category TEXT,
    quantity INTEGER,
    unit_price REAL,
    total_price REAL,
    sale_date TEXT,
    customer_id INTEGER,
    region TEXT
)
""")


# Sample data
```

```

sample_data = [
    (1, 101, 'Laptop', 'Electronics', 2, 900.00, 1800.00, '2025-05-01', 201, 'North'),
    (2, 102, 'Headphones', 'Electronics', 5, 50.00, 250.00, '2025-05-01', 202, 'South'),
    (3, 103, 'Chair', 'Furniture', 4, 70.00, 280.00, '2025-05-02', 203, 'East'),
    (4, 101, 'Laptop', 'Electronics', 1, 900.00, 900.00, '2025-05-03', 204, 'West'),
    (5, 104, 'Notebook', 'Stationery', 10, 3.00, 30.00, '2025-05-03', 205, 'North'),
    (6, 105, 'Pen', 'Stationery', 15, 1.50, 22.50, '2025-05-04', 206, 'South'),
    (7, 106, 'Table', 'Furniture', 2, 150.00, 300.00, '2025-05-04', 207, 'East'),
    (8, 102, 'Headphones', 'Electronics', 3, 50.00, 150.00, '2025-05-05', 208, 'West'),
    (9, 103, 'Chair', 'Furniture', 2, 70.00, 140.00, '2025-05-06', 209, 'North'),
    (10, 107, 'Mouse', 'Electronics', 4, 25.00, 100.00, '2025-05-06', 210, 'South')
]

```

```

cursor.executemany("""
INSERT INTO sales (
    sale_id, product_id, product_name, category, quantity,
    unit_price, total_price, sale_date, customer_id, region
) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
""", sample_data)

```

```

conn.commit()

```

```

# Query the full table

```

```

df = pd.read_sql_query("SELECT * FROM sales", conn)

```

```

# Display the table

```

```

print(df)

```

```
print(df.head()) # Shows the first 5 rows
```

```
summary = pd.read_sql_query("""  
    SELECT category, SUM(quantity) AS total_quantity, SUM(total_price) AS  
    total_revenue  
    FROM sales  
    GROUP BY category  
""", conn)
```

```
print(summary)
```

```
# 1. Run SQL to get total quantity and total revenue
```

```
query = """  
SELECT product_name,  
    SUM(quantity) AS total_quantity,  
    SUM(total_price) AS total_revenue  
FROM sales  
GROUP BY product_name  
"""
```

```
df_summary = pd.read_sql_query(query, conn)
```

```
# 2. Print simple sales info
```

```
print("=== Simple Sales Summary ===")
```

```
for index, row in df_summary.iterrows():
```

```
    print(f"{row['product_name']}: Quantity Sold = {row['total_quantity']}, Revenue =  
    ${row['total_revenue']:.2f}")
```

```
# 3. Plot bar chart for total revenue
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(df_summary['product_name'], df_summary['total_revenue'], color='skyblue')
```

```
plt.title('Total Revenue by Product')
```

```
plt.xlabel('Product Name')
```

```
plt.ylabel('Revenue ($)')
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# 4 Revenue by date
```

```
df_date = pd.read_sql_query("""
```

```
    SELECT sale_date, SUM(total_price) AS total_revenue
```

```
    FROM sales
```

```
    GROUP BY sale_date
```

```
    ORDER BY sale_date
```

```
""", conn)
```

```
plt.figure(figsize=(8, 5))
```

```
plt.plot(df_date['sale_date'], df_date['total_revenue'], marker='o', linestyle='-',  
color='green')
```

```
plt.title('Revenue Over Time')
```

```
plt.xlabel('Sale Date')
```

```
plt.ylabel('Total Revenue ($)')
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

5 Pie Chart – Revenue Distribution by Category

```
df_cat = pd.read_sql_query("""
    SELECT category, SUM(total_price) AS revenue
    FROM sales
    GROUP BY category
""", conn)

plt.figure(figsize=(6, 6))

plt.pie(df_cat['revenue'], labels=df_cat['category'], autopct='%1.1f%%',
startangle=140)

plt.title('Revenue Distribution by Category')

plt.axis('equal')

plt.show()
```

6 Horizontal Bar Chart

```
df_region = pd.read_sql_query("""
    SELECT region, SUM(quantity) AS total_quantity
    FROM sales
    GROUP BY region
""", conn)

plt.figure(figsize=(7, 5))

plt.barh(df_region['region'], df_region['total_quantity'], color='orange')

plt.title('Quantity Sold by Region')

plt.xlabel('Total Quantity')
```

```
plt.ylabel('Region')
```

```
plt.tight_layout()
```

```
plt.show()
```

7 Pivot the data for stacked bar

```
df_stack = pd.read_sql_query("""
    SELECT product_name, region, SUM(quantity) AS quantity
    FROM sales
    GROUP BY product_name, region
""", conn)
```

```
pivot_df = df_stack.pivot(index='product_name', columns='region',
values='quantity').fillna(0)
```

```
pivot_df.plot(kind='bar', stacked=True, figsize=(10, 6), colormap='tab20')
plt.title('Quantity Sold by Product and Region')
plt.xlabel('Product Name')
plt.ylabel('Quantity')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Output:

	sale_id	product_id	product_name	category	quantity	unit_price	total_price	sale_date	customer_id	region
0	1	101	Laptop	Electronics	2	900.0	1800.0	2025-05-01	201	North

1	2	102	Headphones	Electronics	5	50.0	250.0	2025-05-01	
202	South								
2	3	103	Chair	Furniture	4	70.0	280.0	2025-05-02	203
	East								
3	4	101	Laptop	Electronics	1	900.0	900.0	2025-05-03	
204	West								
4	5	104	Notebook	Stationery	10	3.0	30.0	2025-05-03	
205	North								
5	6	105	Pen	Stationery	15	1.5	22.5	2025-05-04	206
	South								
6	7	106	Table	Furniture	2	150.0	300.0	2025-05-04	207
	East								
7	8	102	Headphones	Electronics	3	50.0	150.0	2025-05-05	
208	West								
8	9	103	Chair	Furniture	2	70.0	140.0	2025-05-06	209
	North								
9	10	107	Mouse	Electronics	4	25.0	100.0	2025-05-06	
210	South								

sale_id	product_id	product_name	category	quantity	unit_price	total_price
sale_date	customer_id	region				

0	1	101	Laptop	Electronics	2	900.0	1800.0	2025-05-01	
201	North								
1	2	102	Headphones	Electronics	5	50.0	250.0	2025-05-01	
202	South								
2	3	103	Chair	Furniture	4	70.0	280.0	2025-05-02	203
	East								
3	4	101	Laptop	Electronics	1	900.0	900.0	2025-05-03	
204	West								
4	5	104	Notebook	Stationery	10	3.0	30.0	2025-05-03	
205	North								

category	total_quantity	total_revenue
0 Electronics	15	3200.0

1	Furniture	8	720.0
2	Stationery	25	52.5

=== Simple Sales Summary ===

Chair: Quantity Sold = 6, Revenue = \$420.00

Headphones: Quantity Sold = 8, Revenue = \$400.00

Laptop: Quantity Sold = 3, Revenue = \$2700.00

Mouse: Quantity Sold = 4, Revenue = \$100.00

Notebook: Quantity Sold = 10, Revenue = \$30.00

Pen: Quantity Sold = 15, Revenue = \$22.50

Table: Quantity Sold = 2, Revenue = \$300.00





